QUESTION 1

Arrange the following steps in the order in which a CPU would need to perform them to fetch data from memory:

1. 1. latch the resulting data into a register
2. 2. generate the target address
3. 3. apply the address to the address bus
4. 4. apply a read-enable signal

Note: not all CPUs work in exactly this fashion. Certain signals may be applied simultaneously, or in some cases read-enable may be implicit and not even a separate signal. However, as a logical flow, these are the steps that need to be followed.

Answer is Wrong **Score: 0 / 1**

Options

Wrong

A. 1, 2, 3, 4

B. 2, 3, 1, 4

Expected

C. 2, 3, 4, 1

D. 3, 1, 2, 4

FEEDBACK

Explanation:

To fetch data from memory, a CPU follows a specific sequence of steps:

Generate the target address: The CPU must first determine the address of the data it needs to fetch.
Apply the address to the address bus: The CPU then places this address on the address bus, which connects the CPU to memory.
Apply a read-enable signal: Next, the CPU sends a read-enable signal to the memory to indicate that it wants to read data from the specified address.
Latch the resulting data into a register: Finally, once the data is available on the data bus, the CPU latches it into a register for further processing.

This sequence ensures that the CPU correctly fetches data from the memory location it needs.

QUESTION 2

A "bus" in a microcontroller is:

Not Attempted **Score: 0 / 1**

Options

A. A thick wire used to carry a large amount of current

B. A moving part associated with the MCU

Expected

C. A collection of related wires

D. An external power source for the MCU

FEEDBACK

Explanation:
In microcontroller terminology, a "bus" refers to a set of electrical pathways used for communication between the various components of the microcontroller. These pathways are composed of multiple wires that work together to transfer data, address signals, and control signals among the CPU, memory, and peripheral devices. This coordinated system of wires ensures efficient and organized communication within the microcontroller.

Options A and B describe unrelated concepts, and option D is incorrect as it describes a power source rather than a communication pathway.

QUESTION 3

A CPU has 128KB of SRAM connected to its starting at address 0x00300000. What is the address of the last byte in the SRAM?

Not Attempted **Score: 0 / 1**

Options

Expected

A. 0x0031FFFF

B. 0x00301FFF

C. 0x0031FFFE

D. 0x002FFFFF

FEEDBACK

Explanation:
To find the address of the last byte in the 128KB SRAM starting at address 0x00300000, we need to follow these steps:

1. Convert 128KB to bytes:

128KB = 128 * 1024 bytes
128KB = 131072 bytes

2.Identify the starting address:

The starting address given is 0x00300000.

3.Calculate the ending address:

The ending address is calculated by adding the size of the SRAM to the starting address and then subtracting 1 (because the starting address is inclusive).
Ending address = (Starting address) + (Size in bytes) - 1

4.Perform the calculation:
Starting address in hexadecimal: 0x00300000
Size in bytes: 131072
Convert 0x00300000 to decimal: 0x00300000 = 3145728
Add the size in bytes to the starting address: 3145728 + 131072 = 3276800
Subtract 1 from the result to get the address of the last byte: 3276800 - 1 = 3276799
Convert 3276799 back to hexadecimal: 3276799 = 0x0031FFFF

Thus, the address of the last byte in the 128KB SRAM is 0x0031FFFF, which corresponds to option A.

QUESTION 4

Common Statement for Questions: 4 to 5

For the following Questions, Read from pg. 145 of RM0490 Reference manual.

Reference manual Link: Link

What will happen if both pull-up and pull-down resistors are connected to a GPIO port?

Not Attempted **Score: 0 / 1**

Options

A. The GPIO pin will exhibit undefined behavior

B. The voltage at the GPIO pin will stabilize at Vdd/2

Expected

C. This will cause a continuous flow of current through the resistors from supply to ground

D. The CPU will be damaged

FEEDBACK

Explanation: When both pull-up and pull-down resistors are connected to a GPIO pin simultaneously, they create a direct path between the supply voltage (Vdd) and ground (GND) through these resistors. This results in a continuous flow of current through the resistors, wasting power and potentially causing excessive current draw from the power supply. This configuration is not recommended as it can lead to increased power

consumption and unnecessary heating of components. Therefore, option C correctly describes the consequence of having both pull-up and pull-down resistors connected to a GPIO port.

QUESTION 5

Common Statement for Questions: 4 to 5

For the following Questions, Read from pg. 145 of RM0490 Reference manual.

Reference manual Link: Link

What should be the bit value of the mode in the GPIO port mode register (GPIOx_MODER) to set the port pin for output?

Not Attempted **Score: 0 / 1**

Options

A. 00

Expected

B. 01

C. 10

D. 11

FEEDBACK

Explanation:

The GPIO port mode register (GPIOx_MODER) in STM32 microcontrollers determines the mode of each GPIO pin. Each pin is configured using two bits. The possible configurations are as follows:

00: Input mode
01: Output mode
10: Alternate function mode
11: Analog mode

To set a port pin for output, the mode value should be 01. Therefore, the correct answer is option B) 01.

QUESTION 6

Common Statement for Questions: 6 to 7

CPU has supply voltage of 3.3V. You use a pullup resistor of 1000 ohms on a GPIO port
How much current flows through the resistor when the input is not connected?

Not Attempted **Score: 0 / 1**

Options

A. 1

Expected

B. 0

C. 3.3

D. 2

FEEDBACK

The whole point of pullup is the following: because the input is not connected, there is no place for the current to flow, so I must be 0. This will mean that $V_{GPIO} = Vdd_{pull-up}$

$$I = RV_{pull-up} - V_{GPIOpin}$$

Where:

$V_{pull-up}$ is the voltage supplied to the pull-up resistor (3.3V).

$V_{\text{GPIO pin}}$ pin is the voltage at the GPIO pin (3.3V).

R is the resistance of the pull-up resistor (1000 ohms).

The GPIO pin is at 3.3V, then the current I would be:

$$I = \frac{3.3V - 3.3V}{1000\Omega} = 0 \text{ A}$$

Since I = 0 A, this means there is no current flowing through the resistor when the GPIO pin is at 3.3V.

QUESTION 7

Common Statement for Questions: 6 to 7

CPU has supply voltage of 3.3V. You use a pullup resistor of 1000 ohms on a GPIO port
How much power is dissipated in the resistor when the input is connected to ground?

Not Attempted **Score: 0 / 1**

Options

A. 0.033 mW

B. 0 W

Expected

C. 10.89 mW

D. 3.3 mW

FEEDBACK

Explanation

Calculate the current through the resistor:

When the GPIO port is connected to ground, the full supply voltage (3.3V) is across the resistor. Using Ohm's Law:

$$-I = \frac{V}{R}$$

where:

$$V = 3.3\,\mathrm{V}$$
$$R = 1000\,\Omega$$

Substituting the values:

$$I = \frac{3.3\,\mathrm{V}}{1000\,\Omega} = 0.0033\,\mathrm{A} = 3.3\,\mathrm{mA}$$

Calculate the power dissipated in the resistor:

The power dissipated in a resistor can be calculated using the formula:

$$P = V \times I$$

We already have $V = 3.3\,\mathrm{V}$ and $I = 0.0033\,\mathrm{A}$:

$$P = 3.3\,\mathrm{V} \times 0.0033\,\mathrm{A} = 0.01089\,\mathrm{W} = 10.89\,\mathrm{mW}$$

Therefore, the power dissipated in the resistor when the input is connected to ground is **10.89 mW**.

QUESTION 8

HAL in embedded systems refers to:

Not Attempted **Score: 0 / 1**

Options

A. Hindustan Aeronautics Limited

Expected

B. Hardware Abstraction Layer

C. High-level Assembly Language

D. High Availability Load-balancing

FEEDBACK

Explanation:

HAL in embedded systems stands for Hardware Abstraction Layer.
The Hardware Abstraction Layer (HAL) is a software layer that provides a consistent interface to the hardware components of a system.
It abstracts low-level hardware details and complexities, allowing application software to interact with the hardware in a standardized manner.
Option A, Hindustan Aeronautics Limited, is incorrect as it refers to a different entity.
Options C and D are unrelated to the term HAL in embedded systems.

Therefore, the correct answer is B) Hardware Abstraction Layer, which is a fundamental concept in embedded systems design and programming.

QUESTION 9

Given the following Arduino code snippet, what will be the state of the LED connected to pin 13 after executing the loop function twice?

```
#define LED 13

void setup() {
    pinMode(LED, OUTPUT);
}

void loop() {
    digitalWrite(LED, HIGH);
    delay(500);
    digitalWrite(LED, LOW);
    delay(500);
}
```

Not Attempted **Score: 0 / 1**

Options

A. The LED will be on

Expected

B. The LED will be off

C. The LED will blink once and remain off

D. The LED will blink twice and remain off

FEEDBACK

Explanaation:

In the **loop()** function:

digitalWrite(LED, HIGH); turns the LED on.
delay(500); keeps the LED on for 500 milliseconds.
digitalWrite(LED, LOW); turns the LED off.
delay(500); keeps the LED off for 500 milliseconds.

The loop() function repeats indefinitely. After executing the loop() function twice:

In the first iteration, the LED will be on for 500 ms and then off for 500 ms.
In the second iteration, the LED will again be on for 500 ms and then off for 500 ms.

Thus, after two iterations, the LED will be off at the end of the second delay(500).

QUESTION 10

What is the primary purpose of general-purpose I/O (GPIO) pins on a microcontroller?

Not Attempted **Score: 0 / 1**

Options

A. To provide power to the microcontroller

B. To perform arithmetic operations

Expected

C. To interface with external devices

D. To store data

FEEDBACK

Explanation: The primary purpose of general-purpose I/O (GPIO) pins on a microcontroller is to interface with external devices. These pins can be configured as either inputs or outputs, allowing the microcontroller to read signals from sensors, switches, and other devices, or to control actuators, LEDs, and other peripherals.

QUESTION 11

Which of the following is NOT a function of a GPIO pin?

Not Attempted **Score: 0 / 1**

Options

A. Detecting digital signals

Expected

B. Performing arithmetic operations

C. Acting as an actuator

D. Interfacing with sensors

FEEDBACK

Explanation: A GPIO (General-Purpose Input/Output) pin is used for interfacing with digital signals, sensors, and actuators. It does not perform arithmetic operations, which are tasks handled by the processor or dedicated arithmetic units within a microcontroller or computer.

QUESTION 12

Consider a dataframe with 8 bits represented as 111X0101, where X is a corrupted bit. The dataframe uses even parity for error detection. What should be the value of X?

Not Attempted **Score: 0 / 1**

Your answer

No answer given

Expected answer

Expected answer: 1

FEEDBACK

Explanation:

To determine the value of the corrupted bit X in 111X0101 using even parity:

1. Even parity requires the total number of 1s to be even.
2. Count the 1s in 111X0101 excluding X: there are 5 ones.
3. To make the total even, X must be 1 (since 5 is odd, adding 1 will make it 6, which is even).

Therefore, the value of X should be 1.